Radhe Patel

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Education

Purdue University, College of Engineering — ***West Lafayette, IN***
First-Year Engineering (intended major: Electrical Engineering) — Expected May 2030
Pursuing Bachelor of Science | GPA: 4.0/4.0
Relevant Coursework: Calculus II, C Programming (CS 159), ENGR 131, EPICS

Ivy Tech Community College — ***Indiana***
Indiana College Core; 30 college credits completed during high school — Aug 2024 - May 2026

Engineering Experience

Purdue Lunabotics — ***West Lafayette, IN***
Printed Circuit Board Designer — Sep 2026 - Present
- Design printed circuit boards for the team's autonomous lunar mining robot.

Purdue EPICS Assistive Technology Team — ***West Lafayette, IN***
Design Lead — Aug 2026 - Present
- Lead design of a magnetic hand-guidance writing device for children with cerebral palsy; translate community partner needs into engineering requirements on a six-person team.

Compal Electronics — ***Logansport, IN***
Mechanical Engineering Intern — Aug 2025 - Jan 2026
- Designed a chip-support structure for a medical refrigeration system in Fusion 360; delivered 3D models to specified dimensions within an engineering design workflow.

Electrical and Embedded Projects

4S Li-ion Battery Management System | *C, Python, KiCad* — 2026
- Fitted a 2-RC cell model to published McMaster lab data; achieved 0.96-1.78% state-of-charge RMSE with an Extended Kalman Filter across four drive cycles at 25 C.
- Implemented C99 protection and balancing with 24 unit checks and six software-in-the-loop fault checks; designed a TI BQ76920 analog front-end schematic in KiCad.

800 V SiC Inverter and Gate Driver | *ngspice, Python, KiCad* — 2026
- Modeled a 1200 V SiC MOSFET and simulated double-pulse testing; selected gate resistors, Miller clamp, and DESAT protection for a TI UCC21750-Q1 gate-driver schematic.
- Ran 5,000-sample Monte Carlo thermal studies; selecting 10 kHz at peak torque lowered simulated cases above 150 C from 24.4% to 1.6%.

tinyspice Circuit Simulator | *C, Python, ngspice* — 2026
- Built a SPICE-style simulator in C using Modified Nodal Analysis, Newton-Raphson iteration, LU factorization, and trapezoidal integration for nonlinear transient circuits.
- Validated against analytical solutions and ngspice: 0.009% normalized RMS difference for a 48 V-to-12 V buck converter; added 36 numerical and device-model checks.

Crow Embedded Alarm Controller | *ESP32, Arduino, BLE* — 2026
- Developed ESP32 firmware and a browser simulator for a BLE-linked alarm controller; added 179 automated checks covering HMAC messaging, state transitions, and fault scenarios.

Technical Skills

- **Design:** KiCad schematic capture, PCB design, Fusion 360, SolidWorks
- **Analysis:** ngspice / SPICE, MATLAB / Simulink, NumPy / SciPy, fault injection, Monte Carlo analysis
- **Programming and tools:** C / C++, Python, JavaScript / TypeScript, Git, Microsoft Excel

Additional Software Projects

StudyMon: Offline C and Calculus study RPG with in-browser Clang compilation and automated grading.
FitPip: Offline fitness tracking and device sync. **Catching Days**: Calendar-aware study planner and scheduling.